

House Price Prediction — Summary of Findings

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Overview

This project focuses on predicting house prices using machine learning techniques. The dataset was loaded using Pandas and explored to understand its features and overall structure. After checking for missing values and duplicate records, the data was preprocessed by converting categorical variables into numerical form using one-hot encoding. The cleaned dataset was then used to train and evaluate predictive models.

Models Used

Two regression models were implemented:

- Linear Regression
- Random Forest Regressor

Model Performance

The dataset was divided into training and testing sets using an 80:20 split. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

The Linear Regression model explains approximately 65% of the variance in house prices ($R^2 = 0.65$), while the remaining variance may be attributed to factors not captured in the dataset, such as location-specific premiums, property age, or market conditions. Additionally, the MAE represents roughly 8–10% of the average house price in this dataset, indicating reasonably accurate but not highly precise predictions.

Model	MAE (₹)	RMSE (₹)	R^2 Score
Linear Regression	9,70,043	13,24,507	0.6529
Random Forest	10,21,546	14,00,566	0.6119

Key Findings

The analysis showed that area is the most influential factor affecting house prices. Features such as bathrooms, air conditioning, and parking availability also contribute significantly to the final price. Data visualizations, including a histogram, correlation heatmap, actual vs. predicted plot, and feature importance chart, helped identify these relationships.

Among the two models tested, Linear Regression performed better than Random Forest on this dataset by achieving a higher R^2 score and lower prediction errors. This suggests that the relationship between the available features and house prices is reasonably linear.

Conclusion

Overall, the project successfully demonstrated the complete machine learning workflow, including data loading, preprocessing, visualization, model building, and evaluation. The results indicate that machine learning can provide reliable estimates of house prices and can support real estate businesses in making informed pricing decisions. Focusing on important features such as property area and key amenities can further improve pricing strategies.